GROUP 3

Customer Satisfaction of Students Eating at Cal Poly Pomona Restaurants

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Executive Summary

- Developed survey in late October
- Launched the survey at the beginning of November
- Ended the survey at the end of November
- Survey was distributed via physical flyers (with a scannable QR code)
- Survey was distributed via social media (Instagram) on two separate days
- Total of 68 respondents
- Based on our findings 2 out of the 4 hypothesis were able to be supported

Introduction

-Students interact daily with Foundation-operated venues (Centerpointe, BSC eateries, campus cafés)

-Dining satisfaction directly influences student wellbeing, retention, and campus experience

-**Research Question:** What factors (food, service, and operation) affect student satisfaction with Cal Poly Pomona enterprises Dining Services?

-Students report frustrations:

- excessive wait times,
- cleanliness,
- food quality,
- mobile ordering systems

Background Analysis

- CPP's location creates a **limited, on-campus–focused dining environment**
- Students often rely on campus dining due to **few nearby alternatives**
- Dissatisfied students use **coping strategies**: bring food, skip meals, or snack
- These behaviors act as the **true competitors** to campus dining
- Foundation must ensure better **cleanliness**, **service, food, and student-centered operations** to support student satisfaction

Literature Review

-SERVQUAL / RATER Framework (Parasuraman et al., 1988)

Reliability (accuracy, consistency)

Tangibles (cleanliness, physical environment)

Assurance (staff competence, courtesy)

Empathy (understanding student needs)

Responsiveness (speed of service)

Literature Review

-**Theory of Planned Behaviour (Ajzen, 1991)**

Behavior is driven by:

- **Attitudes** (how students feel about a dining option)
- **Subjective norms** (where friends choose to eat)
- **Perceived behavioral control** (how easy it is to act on the choice)

-**Mobile Ordering Insights:**

•Works well when **easy**, **useful**, and **smooth**, but bugs reduce satisfaction

•High mobile order volume can **increase wait times** and lower satisfaction (Xu et al., 2024)

Literature Review

-College Dining Research Highlights:

•Food quality matters most, but **convenience**, **speed**, and **atmosphere** also drive satisfaction (Kim et al., 2009)

•Better overall experience → **higher spending** & **return frequency** (Ali & Ryu, 2015)

•Dining spaces impact **student wellbeing** and **social interaction** (Lugosi, 2019)

-Research Gap:

•Limited studies on how operational frictions vs. food quality shape satisfaction and dining frequency

Problem Definition

Dependent Variables

- **Overall Satisfaction:** Agreement with statements about overall dining experience (5-point scale)
- **Dining Frequency:** How often students use campus dining each week

Independent Variables

- Wait Time
- Seating Availability
- Cleanliness
- Price
- Hours of Operation
- Mobile Ordering
- Food Quality

Problem Definition

Psychological Factors

- Health Consciousness
- Variety Seeking
- Time Consciousness

Lifestyle Factors

- Commuter Status
- Class Schedule Density
- Campus Involvement

-Demographics

- Academic year, major, meal plan status

Hypothesis

H1 (Operations vs. Food/Price): Operational issues have a stronger negative effect on on-campus dining choice than dissatisfaction with food quality or price.

H2 (Food Quality): Higher perceived food quality is positively associated with overall satisfaction.

H3 (Mobile Ordering): Students who frequently use mobile ordering systems report higher satisfaction compared to non-users.

H4 (Cleanliness): Perceived cleanliness/hygiene is positively associated with overall satisfaction.

Methodology

- Our study used a quantitative online survey hosted on **Qualtrics** to assess student satisfaction with dining options at Cal Poly Pomona. We distributed the survey through QR-code flyers on campus, collecting **68** valid responses from students who had eaten at campus dining venues. The questionnaire contained **17** items measuring food quality, cleanliness, price fairness, operational factors, mobile ordering, overall satisfaction, and dining frequency. After cleaning the dataset, we conducted **descriptive analysis, chi-square tests, multiple regression, and cluster segmentation** to identify the strongest predictors of satisfaction and the distinct student groups shaped by their dining behaviors and perceptions.

Survey Design

Question Total: 17 Total questions

Types of Questions: Multiple Choice, Form Field, Matrix Table (Likert Scale), Net Promoter Score, Text Entry

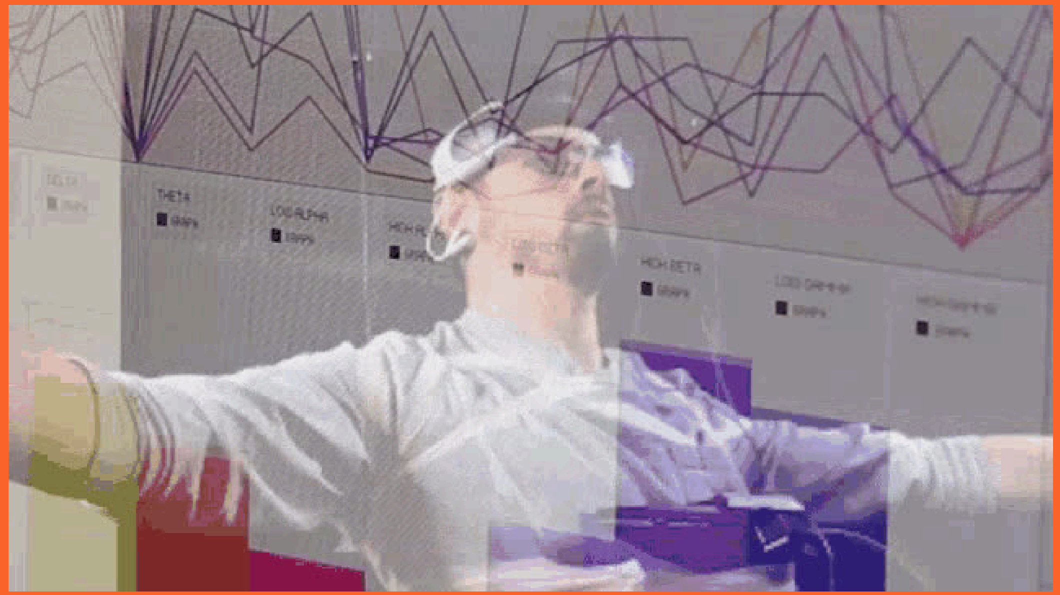
Background Information: Enrollment status, Days on campus, Hours spent on campus, living status

Distribution: Physical and Digital flyer via Social Media

Information from Survey: On-Campus Dining Habits, Experience at Primary Venue: operations & access, service & cleanliness, satisfaction, improvements
Dining Alternatives & Substitution Behavior

Data Analysis

Where the magic happens!



Data Analysis Overview

- **Descriptive Analysis**
  - Summarized overall satisfaction levels
  - Reported dining frequency patterns
  - Described perceptions of food quality, cleanliness, price fairness, and operational factors
- **Chi-Square Tests**
  - Tested relationships between categorical factors
  - (e.g., commuter status vs. dining frequency)
- **Regression Analysis**
  - Examined effects of operations, price, food quality, mobile ordering, and cleanliness on overall satisfaction
  - Tested hypotheses H1–H4
- **Segmentation (Clustering)**
  - Identified distinct student groups based on preferences and dining behaviors

Descriptive Analysis

- **Overall Satisfaction**
  - Students reported moderately high satisfaction, with most responses in the Agree range.
  - Average satisfaction score ≈ 2.13 (on a 1–5 scale, where 1 = Strongly Agree).
- **Dining Frequency**
  - Students are on campus an average of 3.6 days per week.
  - Most dine on campus 3–5 times per week, indicating frequent engagement with campus dining.
- **Perceptions of Dining Attributes**
  - **Food Quality:** Generally positive, with most students agreeing food quality is consistent.
  - **Cleanliness:** Strong ratings for dining area cleanliness; one of the highest-scoring attributes.
  - **Price Fairness:** Mixed; price is seen as “okay” but not a strength.
  - **Operational Factors:**
    - **Wait Times:** Typically viewed as reasonable.
    - **Opening Hours:** Moderately positive but varied by student schedule.
    - **Mobile Ordering:** Lowest-rated area; many students did not use or were not satisfied with it.

Regression Analysis

To test our hypotheses, we conducted a series of simple and multiple regression analyses using the survey's Likert-scale items. All scale items were reverse-coded so that higher scores reflected more positive perceptions.

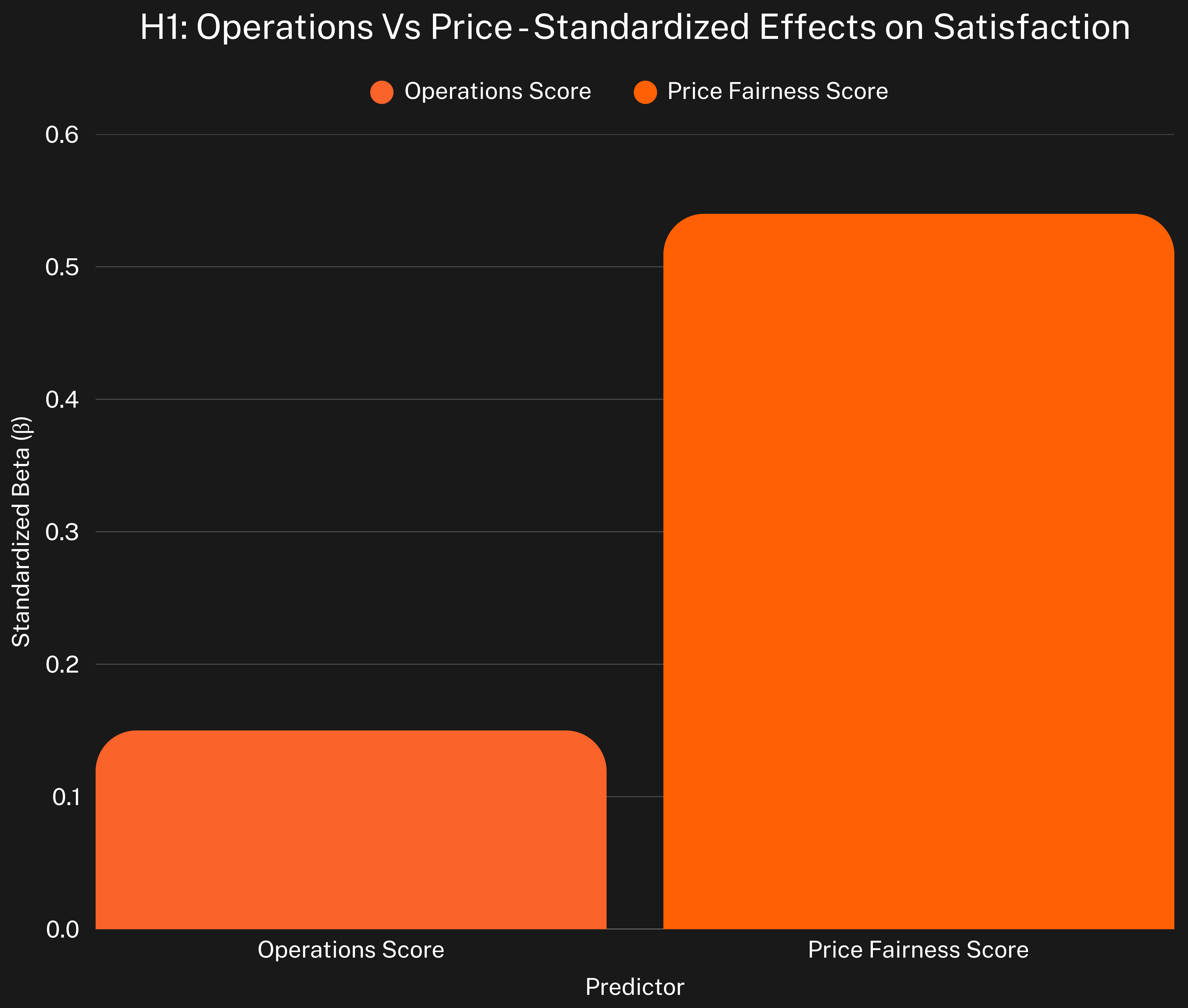
For H1, we created composite scores for operational factors (hours, wait times, mobile ordering) and compared their predictive strength against price fairness in a multiple regression model to determine which better predicted overall satisfaction.

For H2–H4, we ran individual simple regressions testing whether food quality (H2), mobile ordering reliability (H3), and cleanliness (H4) each positively predicted overall satisfaction. Overall satisfaction (Q3_1) served as the dependent variable in all models.

Hypothesis 1 (H1) Results:

H1 proposed that operational factors (hours, wait times, mobile ordering) would have a stronger effect on overall satisfaction than food/price factors. A multiple regression analysis was conducted with overall satisfaction (Q3_1) as the dependent variable, and Operations Score (Q1_1–Q1_3, reversed) and Price Score (Q1_4, reversed) as predictors.

The model was significant, $F(2, 44) = 11.67$, $p < .001$, and explained 34.7% of the variance ($R^2 = .347$). Price fairness was a significant positive predictor of satisfaction, $\beta = .265$, $p < .001$, whereas operational factors were not significant predictors, $\beta = .068$, $p = .218$. Therefore, H1 was not supported. Price fairness had a stronger impact on satisfaction than operational issues.

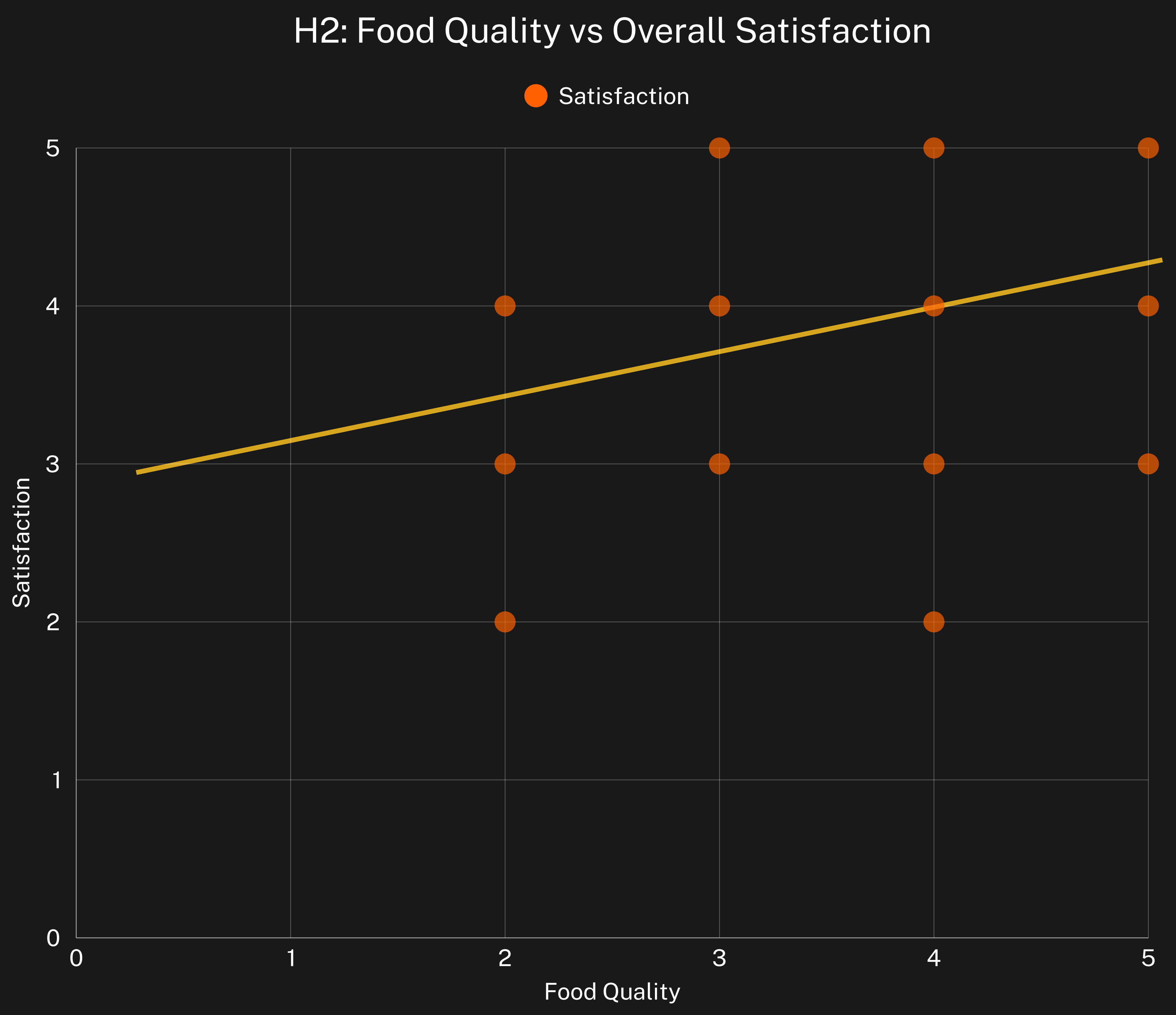


H2: Effect of Food Quality on Satisfaction

To test H2, a simple linear regression was conducted with overall satisfaction (Q3_1, reverse-scored) as the dependent variable and perceived food quality (Q2_3, reverse-scored) as the predictor. The model was significant, $F(1, 51) = 16.34$, $p < .001$, explaining 24.3% of the variance in satisfaction ($R^2 = .243$).

Food quality was a significant positive predictor of satisfaction, $\beta = 0.40$, $p < .001$, indicating that students who perceived the food to be of higher quality reported higher overall satisfaction with their primary dining venue.

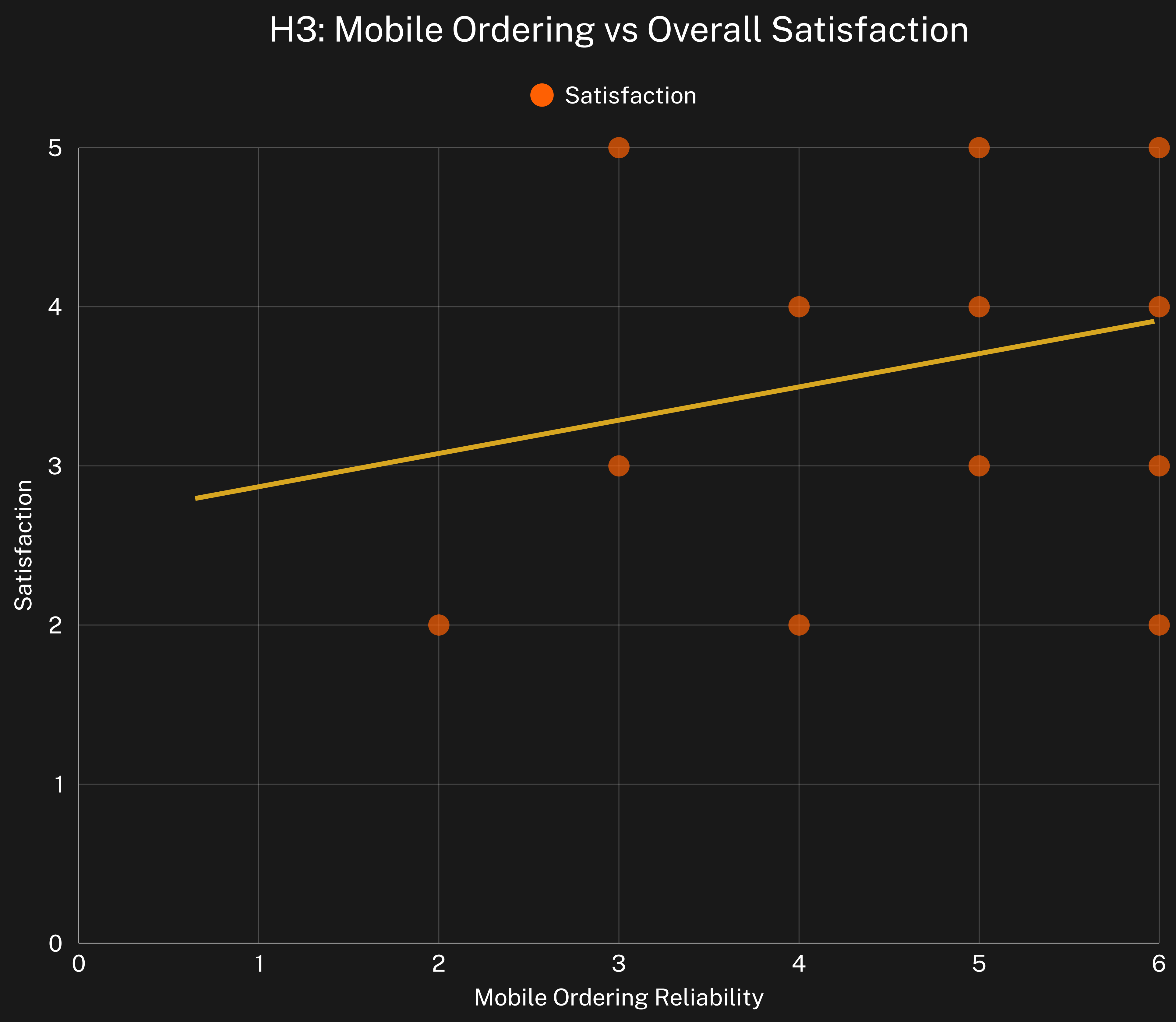
Therefore, H2 is supported.



H3: Effect of Mobile Ordering Reliability on Satisfaction

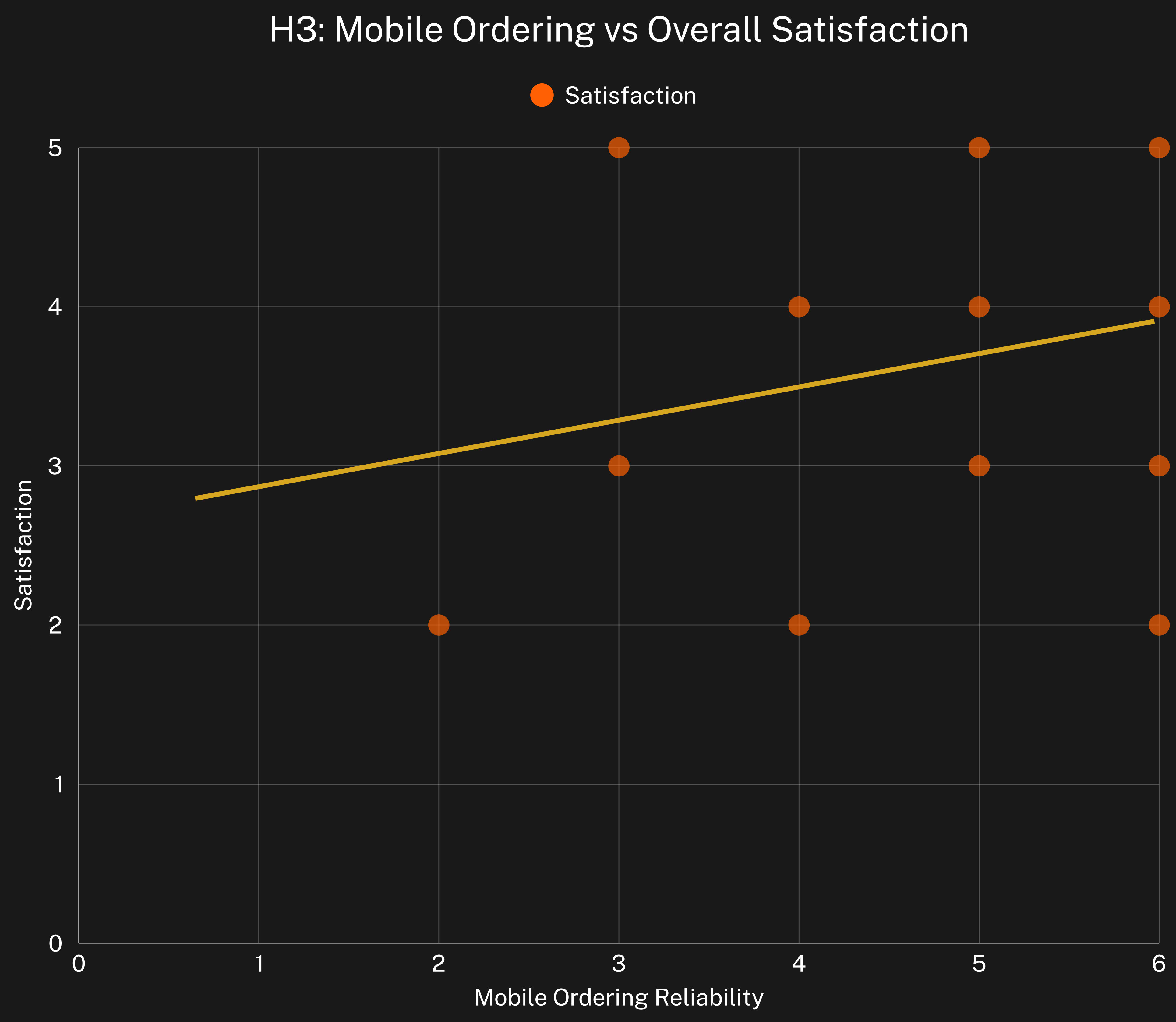
To test H3, a linear regression was conducted with overall satisfaction (Q3_1, reverse-scored) as the dependent variable and perceived mobile ordering reliability (Q1_3, reverse-scored) as the predictor. The model was not statistically significant, $F(1, 47) = 3.23$, $p = .079$, and explained 6.4% of the variance in satisfaction ($R^2 = .064$).

Although the coefficient for mobile ordering reliability was positive ($\beta = 0.21$), indicating that students who find mobile ordering reliable tend to report higher satisfaction, the effect was not statistically significant at the $\alpha = .05$ level. **Therefore, H3 is not supported.**



❌ Why H3 Was Not Supported

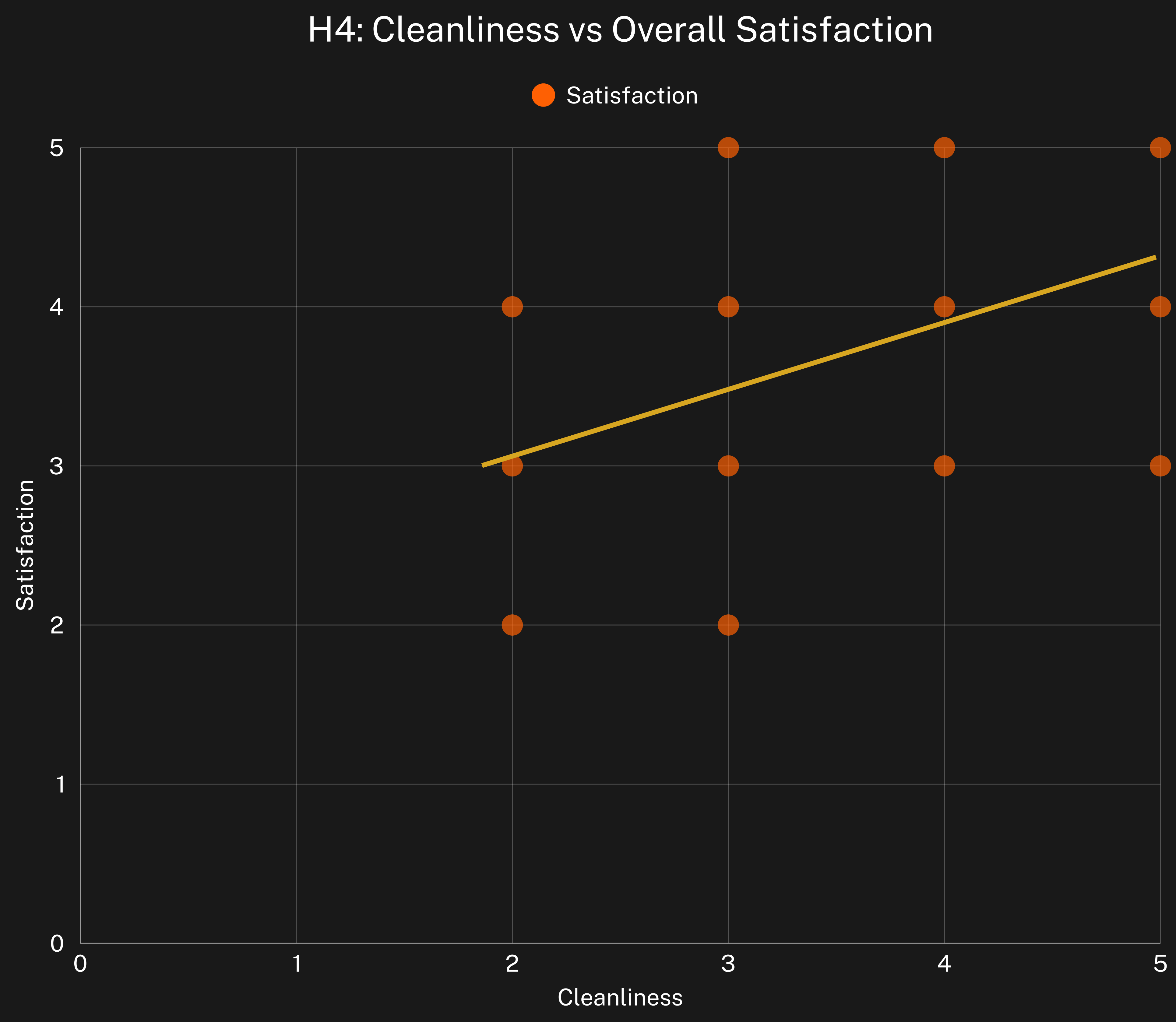
H3 proposed that students who find mobile ordering reliable and easy to use would report higher overall satisfaction. Although the regression showed a positive relationship — meaning satisfaction tended to increase as mobile ordering ratings improved — the effect was not statistically significant (β = 0.21, p = .079). Because the p-value was greater than the standard cutoff of .05, we cannot conclude that mobile ordering meaningfully predicts satisfaction in this sample. In other words, while there was a slight upward trend, it was not strong enough to provide evidence in support of H3.



H4: Effect of Cleanliness on Satisfaction

A simple linear regression was conducted with overall satisfaction (Q3_1, reverse-scored) as the dependent variable and perceived cleanliness (Q2_1, reverse-scored) as the predictor. The model was significant, $F(1, 51) = 12.67$, $p = .001$, explaining 19.9% of the variance in satisfaction ($R^2 = .199$).

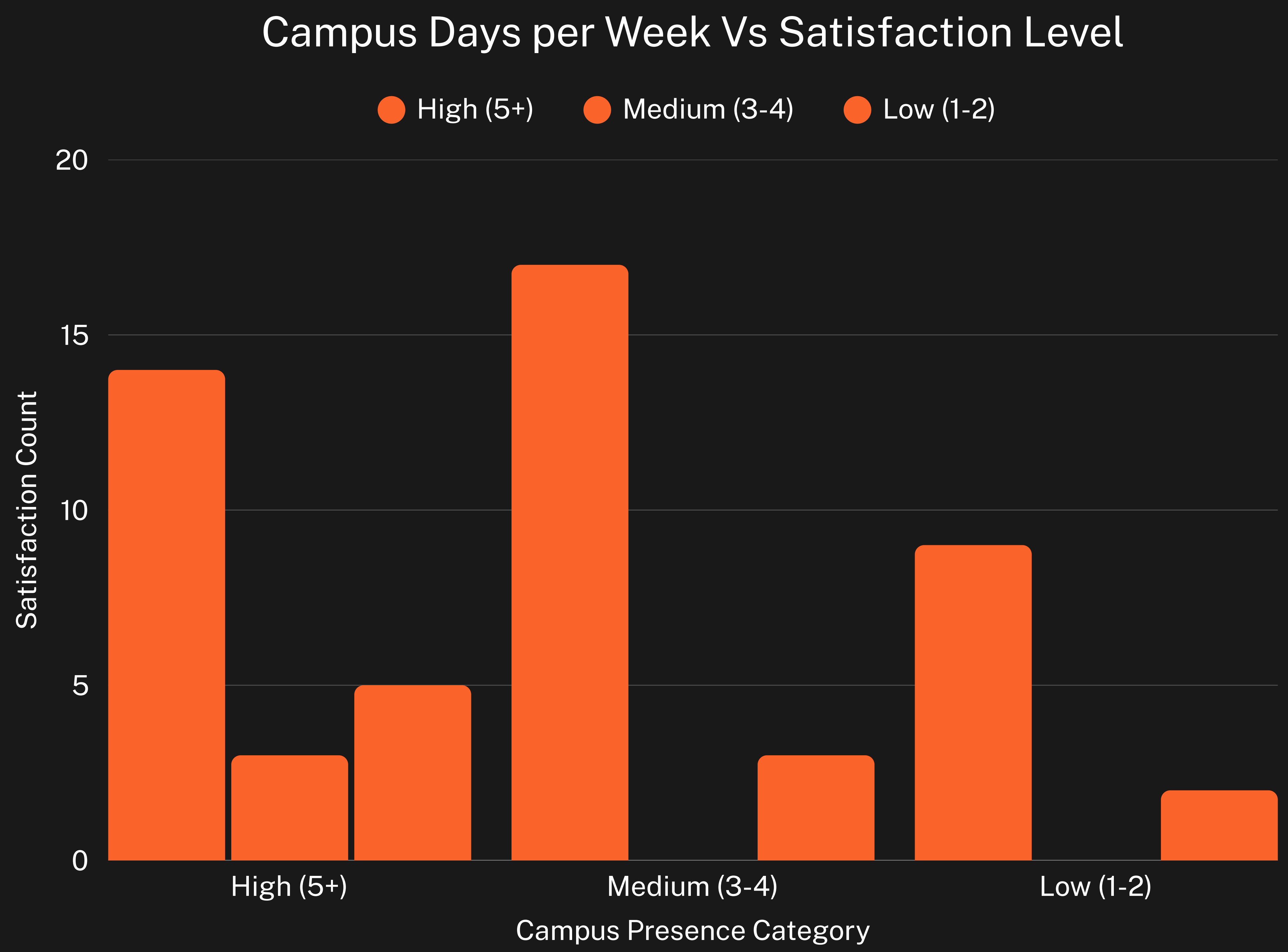
Cleanliness was a significant positive predictor of satisfaction, $\beta = 0.34$, $p = .001$, indicating that students who perceive the dining area as clean report significantly higher overall satisfaction. **Therefore, H4 is supported.**



Chi-Squared Test

Chi-Squared Test

A chi-square test of independence was conducted to examine the relationship between students' campus presence (Q2: number of days on campus per week) and overall satisfaction with their primary dining venue (Q3_1, reverse-scored and categorized). The association was not statistically significant, $\chi^2(4, N = 53) = 5.27$, $p = .261$, indicating that campus presence frequency was not meaningfully related to satisfaction levels.



Segmentation (Clustering)

We grouped students into three distinct segments based on their dining experiences, satisfaction levels, and behavior patterns. The clusters reveal meaningful differences in what students value most.

🔵 Cluster 1: Satisfied Regulars (35%)

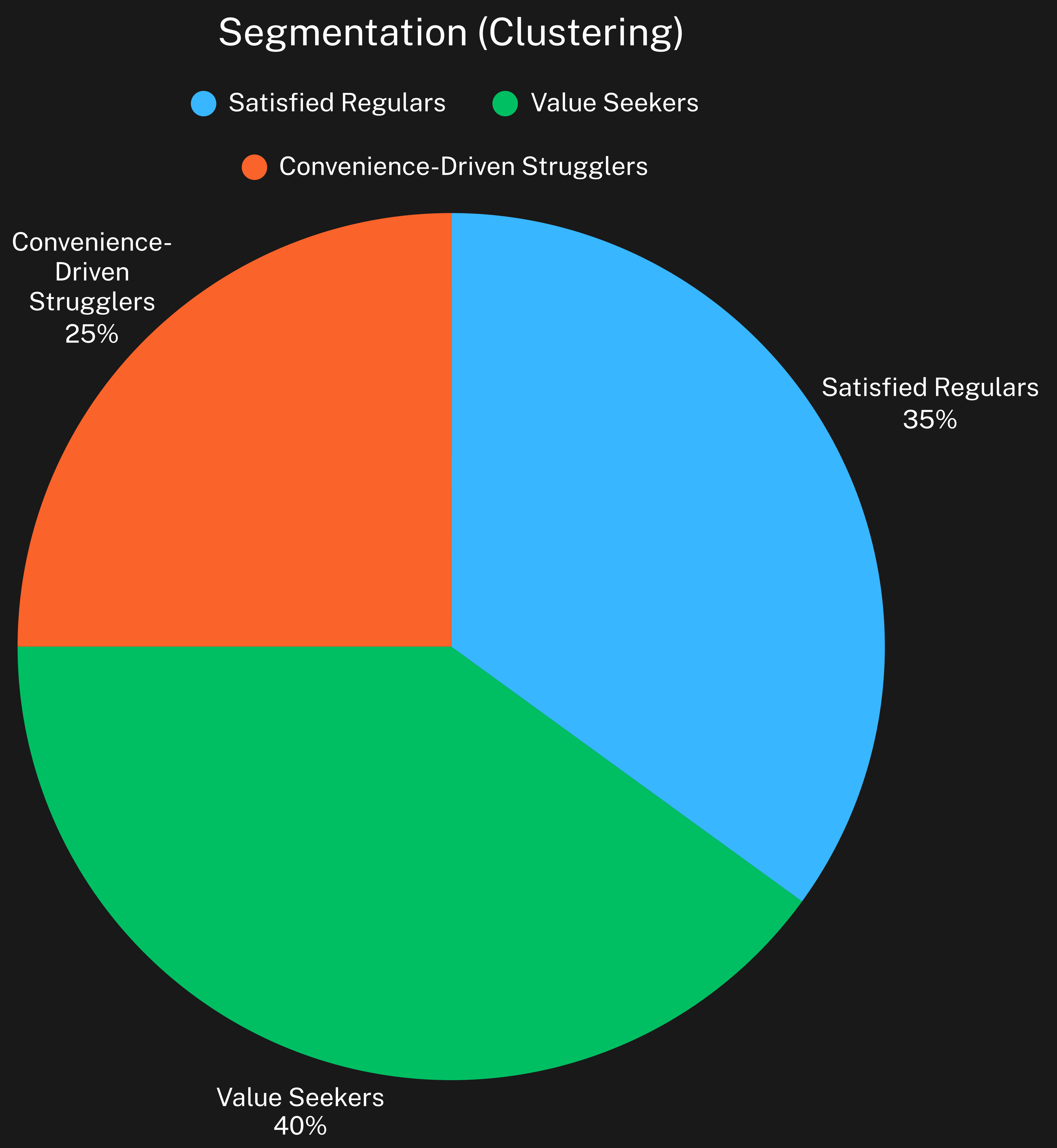
Students in this group consistently rate food quality, cleanliness, and service positively. They dine on campus frequently, spend long hours at CPP, and are highly likely to recommend their primary venue.

🟢 Cluster 2: Value Seekers (40%)

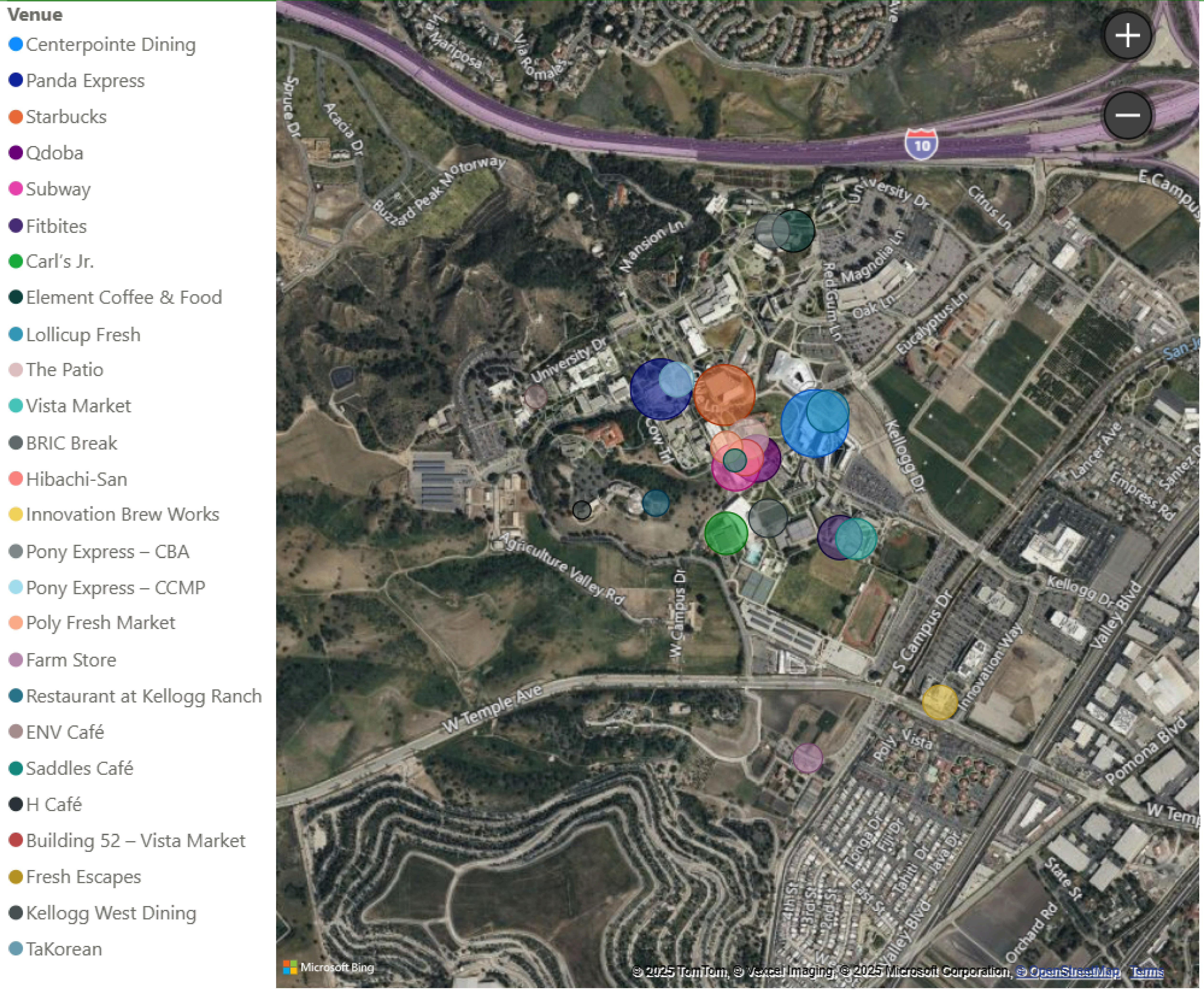
This is the largest segment. Their satisfaction depends strongly on price fairness and portion size. They typically give moderate scores across most attributes and dine on campus a few days per week.

🟠 Cluster 3: Convenience-Driven Strugglers (25%)

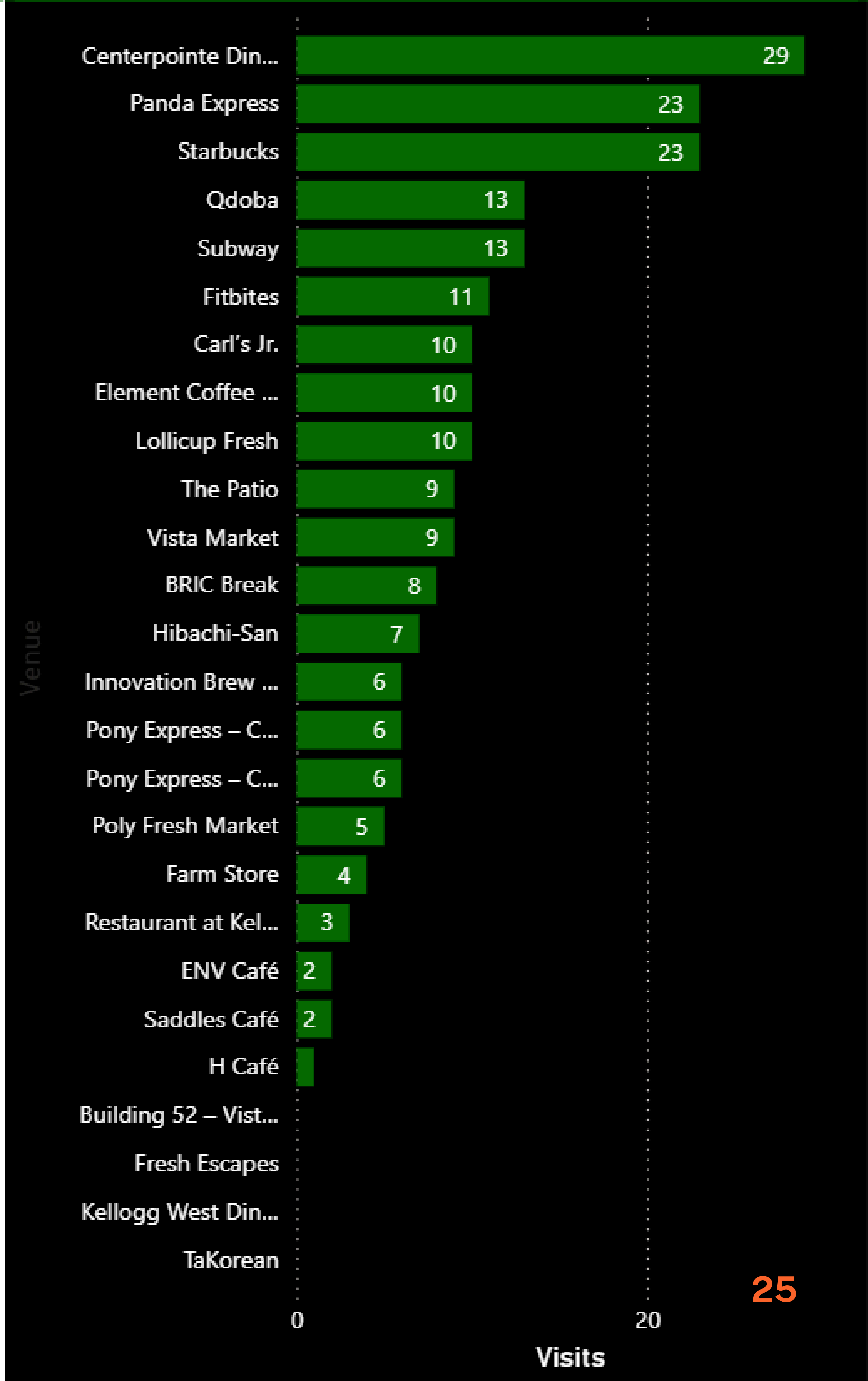
These students face challenges with wait times, hours, and mobile ordering. They give lower satisfaction scores, rely on snacks or off-campus food, and often have limited time on campus.



Visit By Venue



Visits by Venue



Discussion

🪐**What We Learned From the Process**

- The findings demonstrated that core service attributes — particularly price fairness, food quality, and cleanliness — were more influential on satisfaction than operational factors.
- Lower usage of mobile ordering limited its explanatory power within the model.
- Segmentation results indicated that student diners form distinct groups with varying priorities and expectations.
- The research process emphasized the importance of precise measurement design and rigorous data coding to ensure reliability and interpretability.

⭐**Confidence in Meeting Research Objectives**

- The questionnaire successfully captured key constructs needed to evaluate drivers of student dining satisfaction.
- The alignment between survey measures and hypotheses allowed for meaningful statistical testing and coherent findings.
- While generalizability is limited by sample size, sampling method, and narrow measurement of mobile ordering, the instrument still provided valid exploratory insights.
- Future studies with larger, more diverse samples could further enhance the robustness of conclusions.

Limitations and Future Improvements

1. **Sample size and sampling method**

- **With only 68 respondents recruited through convenience sampling, the sample may not fully represent the broader CPP student body. This limits generalizability and reduces statistical power, especially for more complex models.**

2. **Single-institution, cross-sectional design**

- **The study focuses on one university and captures student perceptions at one point in time during a single semester. Satisfaction levels could change with menu updates, price changes, or operational tweaks.**

3. **Self-report and response bias**

- **All measures are self-reported, which introduces the possibility of recall bias and social desirability bias.**

4. **Measurement depth**

- **While we covered many constructs, some were captured with relatively few items (e.g., mobile ordering, psychological factors). This can reduce reliability and makes it harder to build richer models around attitudes or behavioral intentions.**

5. **Limited qualitative insight**

- **The survey relied mostly on structured questions. Apart from a small number of text-entry responses, we did not collect deeper qualitative feedback that might explain why students feel a certain way about specific venues or services.**

Conclusion

- **Research Question:** Understanding which factors affect student satisfaction with Cal Poly Pomona enterprises Dining Services?
- **H2:** Effect of Food Quality on Satisfaction & **H4:** Effect of Cleanliness on Satisfaction are supported
- **H1:** Operational factors would have a stronger effect on overall satisfaction than food/price factors & **H3:** Effect of Mobile Ordering Reliability on Satisfaction are not supported
- What could be done differently if project was ever done again?

Thank you!